A CRISP-DM–Guided Machine Learning Approach to Optimizing High-Impact Journal Publications Using Rapid Miner

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INTRODUCTION

ABSTRACT

This study examines whether measurable paper features can predict a paper's likelihood of being published in a high-impact journal.

WHY ARE WE DOING THIS?

Where a paper is published can significantly influence an academic career. Publishing in high-impact journals affects PhD success, promotions, and a university's reputation. What if researchers knew *before writing* which features to optimize to increase their chances of publishing in a high–impact journal?

RESEARCH QUESTION

Can measurable features of a research paper reliably classify it as belonging to high impact or low impact journals?

DATA

103 peer-reviewed research papers hand-collected from ScienceDirect and other academic databases across AI, Business, Social Sciences, and Healthcare domains.

VARIABLES

Variable	Meaning / Description	Data Type
Impact Factor	Numeric rating of journal	Continuous
Page Count	Total number of pages in the paper	Integer
Word Count	Estimated total word count	Integer
Authors	Number of authors	Integer
Figures	Count of tables and figures	Integer
Paper Type	4 types: review, experimental, conceptual, commentary	Categorical
References	Total number of cited references	Integer

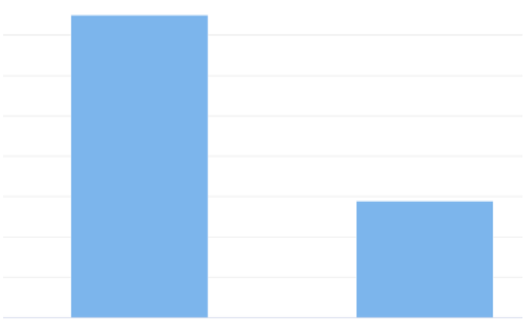
DATA ANALYSIS

CORRELATION MATRIX

Checked for variable correlation > 0.70

Attribu...	figures	pages	words	authors	referen...	impfac...
figures	1	0.400	0.540	0.036	0.211	0.007
pages	0.400	1	0.629	-0.096	0.387	0.029
words	0.540	0.629	1	-0.023	0.547	0.077
authors	0.036	-0.096	-0.023	1	0.186	0.121
references	0.211	0.387	0.547	0.186	1	0.165
impfactor	0.007	0.029	0.077	0.121	0.165	1

CLASS DISTRIBUTION



0 = Low Imp (75) *1 = High Imp (29)*

Papers were classified into high-impact and low-impact journals using an impact factor threshold of 7.

FEATURE ENGINEERING

Created new features:

Visual Density = Figures ÷ Pages
How visually dense the paper looks

Cite Intensity = (Refs ÷ Words)*1000
Number of references per 1000 words

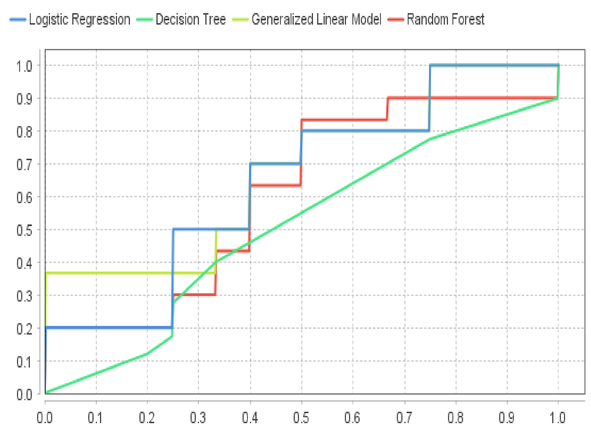
MODELING

METHODOLOGY

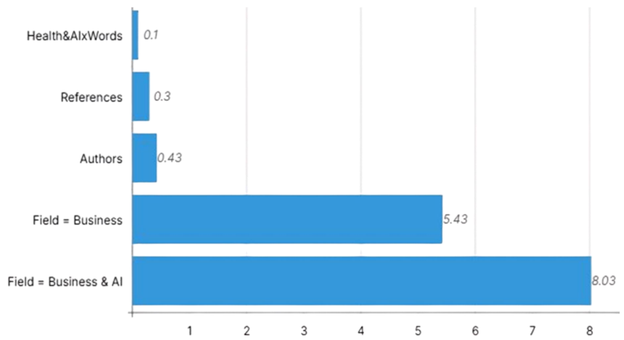
We used multiple models – Generalized Linear Model, Logistic Regression, Decision Tree and Random Forest for comparison on RAPID MINER.

ROC CURVE

Logistic Regression is best model with total time (377ms), specificity (0.96), AUC(0.65) and accuracy (0.70).



MOST IMPORTANT LOG-ODDS



INTERPRETATION

WHAT DO THE RESULTS SAY?

The model is telling us that high-impact journals disproportionately favor business papers (compared to social science and healthcare); even more so when combined with AI. It is always good to add an AI angle to your business papers.

Additionally, more authors slightly increase publication odds and more references have a small positive effect as well. Writing a research paper alone is not the best idea!

Furthermore, our model shows that word counts, number of pages, and figures do not really matter. However, word count does matter when writing a paper in healthcare linked to AI. Over here, more words are always better.

EVALUATION LIMITATIONS

- We evaluated performance using a confusion matrix and found low sensitivity in identifying high-impact papers, largely due to a small, imbalanced dataset.
- The model could have used features like abstract quality, etc. to bring a qualitative aspect to the research.

ACKNOWLEDGEMENT

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